Group 17: Implementation of a Real-Time VoIP Communication System Using TCP and UDP

Project 3

Computer Science 313

Members (Current and Historical Roles)

	Project 1	**Project 2**	**Project 3**
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Courtney Campbell Student Email: 27906124@sun.ac.za	**Role: Client** Client.java LoginController.java ChatApp.java Login.fxml Experiment 1 Experiment 2	**Role: Client** Packet.java Sender.java Experiment 3	**Role: Client** AudioConfig.java AudioSender.java AudioReciever.java AudioMixer.java
Erin Painting Student Email: 27023931@sun.ac.za	**Role: Client** ChatApp.java client.java ChatController.java ChatUI.fxml Message.java README.md	**Role: UI** RecieverController.java SenderController.java MainApp.java ReceiverUI.java SenderUI.java README.md	**Role: UI** ChatController.java ChatUI.fxml WindowManager.java ServerApp.java ServerController.java ServerUI.fxml Experiment 3
Kayla De Beer Student Email: 27480216@sun.ac.za	**Role: Server** Server.java Experiment 3	**Role: Server** RBUDPReceiver.java Experiment 2 Experiment 4	**Role: Server** Server.java Experiment 1

1. Introduction

Voice over Internet Protocol (VoIP) enables voice communication over the Internet by converting audio into digital data packets. This allows users to communicate in real time without relying on traditional telephone networks.

The aim of this project is to develop a VoIP chat system consisting of a central server and multiple clients. The server manages connections, maintains a list of active users, and facilitates the initiation of calls, while voice communication occurs directly between clients using UDP to ensure low latency. In addition to voice calls, the system supports text messaging and voice notes.

The client application includes a graphical user interface (GUI) that allows users to interact with the system, view active users, and initiate communication. This project demonstrates key networking concepts such as client-server architecture, TCP and UDP protocols, and real-time data transmission.

Due to the lack of hardware Acoustic Echo cancellation (AEC), users must wear headphones during calls to prevent acoustic feedback loops.

2. AI Declaration

- Gemini was used to help create the skeleton code for AudioReciever.java, AudioSender.java ,AudioMixer.java and Client.java.
- Gemini was used to understand how audio byte streams work
- Gemini was used to clean comments
- Claude was used to help debug when compiling errors were received
- ChatGPT was used to explain and break down the project and what needed to be implemented
- Claude was used to help create the Makefile
- ChatGPT was used to correct grammar and make the report consistent
- ChatGPT was used to help write a clean README
- Claude was used to help with styling consistency in the UI
- Gemini was used to teach the concepts as well as skeleton for the server side
- Claude was used to help explain how to implement group callings
- ChatGPT was used to confirm correct interpretation of experiment results
- Gemini was used to assist in making the experiment files
- Claude helped with giving a basic UI design and colors
- ChatGPT gave the ideas for the experiments
- Claude helped create downsize sample rate method for experiment 3

3. Experiments

Experiment 1: Effect of Packet Loss on Call Quality (UDP)

- **Hypothesis:** Packet loss will have an immediate and noticeable effect on audio quality, even at the lowest proportion. It will have a noticeable effect at higher proportions, potentially even causing incoherent audio.
- **Method:**
  1. Simulate packet loss by modifying the client to randomly drop packets at proportions {0%, 5%, 10%, 20%, 30%}
  2. Set up a call under each loss level
  3. Rate the audio quality from 1-5
  4. Comment on any delays or gaps in speech, etc.
- **Measurements and Visualisation:**

Packet Loss Rate	Rating (1-5)	Comment
0%	5	Baseline. Clear voice clarity
5%	4	Audio sounds crackly but still understandable
10%	3	Noticeable gaps but mostly understandable
20%	2	Noticeable gaps means only some sentences are understandable
30%	1	Incoherent; very crackly

- **Conclusion:** The hypothesis correctly predicted that the effect will be noticeable at the lowest proportion (5%) and incoherent at the highest proportion (20-30%). The ratings being decent at 5% loss and degrading thereafter is consistent with the extent to which the human brain can fill in gaps. We can also conclude that VoIP is quite sensitive to packet loss and this experiment is very relevant to this application since UDP doesn't guarantee delivery.
- **Limitations:** Although this experiment is valuable in showing the more practical effects of packet loss, the dependent variable is subjective and thus the experiment outcome is slightly less reproducible. The results would still rank similarly (as loss rate increases, rating decreases), but the extent would differ.
- **Potential improvements:** We could offset the effect of the subjective rating by measuring the average rating of more than one person's rating. Alternatively, adding more definitive criteria to the rating could help.

Experiment 2: Effect of Conference Size on P2P Bandwidth

Hypothesis:

In a P2P architecture, the host's upload bandwidth will scale linearly *(O(N)),* increasing by exactly 16KB/s for each participant.

Measurements:

Test different group call sizes using the /conference command:

2 participants (1 target) = 16KB/s

3 participants (2 targets) = 32KB/s

4 participants (3 targets) = 48KB/s

5 participants (4 targets) = 64KB/s

6 participants (5 targets) = 80KB/s

Measure:

Host's continuous upload speed (in KB/s) using a custom network monitor inside AudioSender.java.

Conclusion:

Mathematical proof of O(N) linear bandwidth scaling (exactly 16 KB/s required per additional target). This demonstrates why P2P is not scalable for big group calls, and why big VoIP systems require centralized servers.

Experiment 3: Effect of Sample Rate on perceived voice note quality

Hypothesis

Reducing the audio sample rate will decrease data size and improve transmission efficiency, but will reduce perceived audio quality (clarity, naturalness, and increase distortion)

Method

Three voice notes were recorded at high quality (44,100 Hz) using the same speaker, tone, and sentence content. Each recording type was controlled for vocal intensity. Each recording was then down sampled to 16,000 Hz, 8,000 Hz, and 3,000 Hz for playback testing.

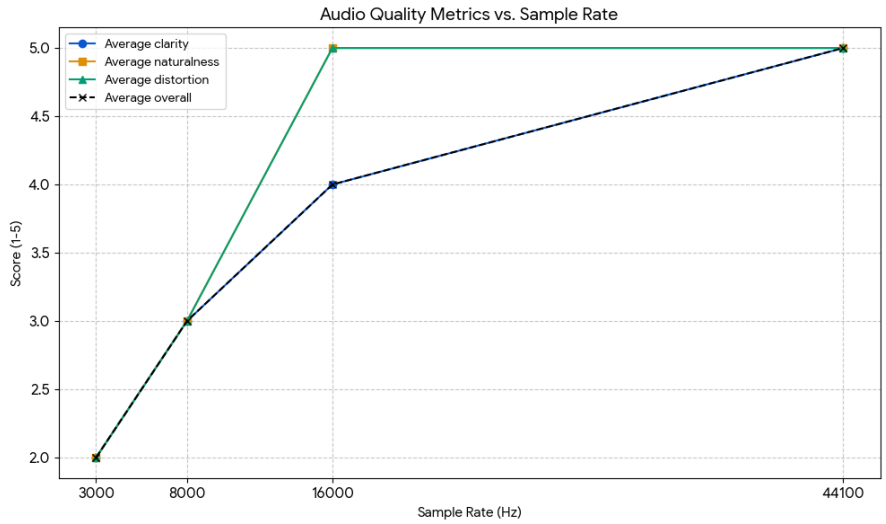
Each playback was evaluated immediately after listening using a 1–5 Likert scale for:

- Clarity: how clear and understandable the speech is
- Naturalness: how realistic and human-like the audio sounds
- Distortion (inverted: 5 = no distortion, 1 = highly distorted): level of audio degradation or artifacts
- Overall quality: general subjective rating combining all factors

Measurements

- Voice Note 1: Loud voice (stable high amplitude speech)
- Voice Note 2: Soft voice (low amplitude speech)
- Voice Note 3: Normal voice (baseline conversational speech)

Sample rate (Hz)	**Average clarity**	**Average naturalness**	**Average distortion**	**Average overall**
44,100	**5**	**5**	**5**	**5**
16,000	**4**	**5**	**5**	**4**
8,000	**3**	**3**	**3**	**3**
3,000	**2**	**2**	**2**	**2**



Conclusion

The results support the hypothesis. As sample rate decreased, perceived audio quality consistently declined across all voice note types.

At 44,100 Hz, audio was fully clear and natural. At 16,000 Hz, quality remained high with only minor clarity loss, aligning with its use in VoIP and voice calls. At 8,000 Hz, degradation became noticeable, producing a more compressed, “telephone-like” sound. At 3,000 Hz, audio quality deteriorated significantly, with reduced clarity and increased distortion, making speech difficult to follow. These results confirm that lower

sample rates reduce data size but introduce a clear trade-off in perceptual quality, with 16,000 Hz appearing to be the lowest acceptable threshold for practical voice communication.

Limitations

- Subjective scoring may introduce bias, even though a fixed scale was used.
- Only one listener was used instead of multiple participants, limiting reliability.
- Environmental noise was not formally measured or controlled.
- Device-specific audio playback hardware may have influenced perceived quality

4. Conclusion

This project successfully implemented a real-time VoIP communication system supporting text messaging, voice notes, and voice calls through a functional GUI. It demonstrated key networking concepts such as client-server coordination, packet-based communication, and real-time audio transmission.

The experiments provided practical insights into system performance. Packet loss was shown to significantly degrade call quality, bandwidth usage increased linearly with the number of participants in group calls, and lower sample rates reduced data size but negatively impacted perceived audio quality. In particular, 16,000 Hz was identified as a reasonable balance between efficiency and clarity for voice communication.

Overall, the project highlights both the effectiveness and limitations of real-time communication systems, emphasizing challenges such as network reliability, scalability, and maintaining audio quality under constraints.